

Business Understanding Report – Heart Disease Prediction

1. Business Problem

Cardiovascular disease is a major health challenge, and early identification of individuals at higher risk can support timely medical intervention. In clinical settings, healthcare professionals assess factors such as age, cholesterol, blood pressure, chest pain, and other patient characteristics when evaluating cardiovascular risk. However, relying solely on individual clinical assessments can make early risk identification difficult, particularly when multiple risk factors interact. The business problem is therefore to use available patient data to identify patterns associated with heart disease and support more consistent risk assessment.

Project Objective

The primary objective of this project is to develop, evaluate, and validate a supervised machine learning classification model capable of accurately predicting patient mortality risk based on clinical features collected during routine patient assessments. By building an accurate risk-scoring pipeline, this project aims to provide healthcare professionals with an actionable decision-support tool that enables early medical intervention, personalized treatment planning, and optimized clinical care pathways.

As a Junior Data Scientist at AnalystLab Africa Consulting, the focus at this stage is not to deploy a diagnostic system, but to transform the raw dataset into a reliable machine-learning-ready format. This involves inspecting and cleaning the data, handling missing values and outliers, engineering relevant features, encoding categorical variables, addressing redundant features, and scaling numerical variables where appropriate.

3. Target Variable

The target variable is **HeartDisease**, which indicates whether heart disease is present. It is a binary classification variable where **1 represents the presence of heart disease and 0 represents its absence**. The remaining patient attributes serve as potential predictor variables. Correctly separating the target from the predictor features is important to ensure that the eventual machine learning model learns from relevant patient characteristics without introducing data leakage.

4. Importance of Predictive Modelling

Predictive modelling can help healthcare organizations identify patients who may be at higher risk based on combinations of observed characteristics. Instead of considering individual risk factors in isolation, a machine learning model can learn patterns across multiple variables and provide a data-driven risk classification. This can support healthcare professionals in prioritizing further assessment and potentially identifying higher-risk patients at an earlier stage.

5. Expected Business Impact

- **Improved Clinical Outcomes:** Early identification of patients at higher risk of heart disease can support timely clinical assessment, closer monitoring, and appropriate intervention, potentially improving patient outcomes.
- **Cost Reduction:** Earlier risk identification may help reduce preventable complications and costly emergency visits or hospital admissions by enabling more proactive patient management.
- **Efficient Resource Allocation:** Predictive insights can help healthcare providers prioritize patients who may require additional consultations, monitoring, or follow-up care, allowing limited healthcare resources to be directed where they are most needed.
- **Data-Driven Decision-Making:** A reliable predictive model can provide healthcare professionals with additional data-driven insights to support, rather than replace, clinical judgment when assessing patient risk.
- **Foundation for Machine Learning:** At this stage, properly cleaning, engineering, and preprocessing the dataset creates a reliable foundation for developing and evaluating a machine learning model. This demonstrates how high-quality data preparation can contribute to effective predictive solutions in healthcare.

Conclusion

The project addresses the need for a structured, data-driven approach to identifying potential heart disease risk. By carefully preparing the dataset through cleaning, feature engineering, encoding, outlier treatment, feature selection, and scaling, the analysis will establish a reliable foundation for subsequent machine learning. The emphasis of this stage is to ensure that preprocessing decisions are technically appropriate, clearly justified, and aligned with the prediction objective.